Demonstrating BloodLink, minute by minute

This guide is your script for the live demo. It follows **one unit of blood** from the moment a patient needs it to the moment it is donated — through every role, every important page and every rule that makes BloodLink more than a register. Each step tells you what to **do**, what to **say**, what to **point at**, and how to answer if a lecturer interrupts. The second half explains **how everything connects** underneath, lists the questions you are likely to be asked with answers, and gives you a recovery plan if anything misbehaves. Every number and message in this guide was checked on the real app.

> THE STORY YOU ARE TELLING
>
> A new member, **Aisha**, registers and files a **Critical Emergency** request: 1 unit of **O+** blood for her relative **Ibrahim Yusuf** at **Barau Dikko Teaching Hospital**, needed within 2 hours. Hospital officer **Maryam** sees it at the top of the Emergency Board, matches a compatible donor (**Kamalu**), **screens** his health on the spot, and records the donation. A second donor (**Grace**) is deferred by the screening. Aisha is told her request is fulfilled and finds the nearest blood bank on the map. The administrator closes the loop with stock, alerts and reports.

Part 1 / Before you present

Get the stage ready

THE NIGHT BEFORE (ABOUT 30 MINUTES)

1. **Latest code.** On the Windows PC, open the project folder and pull `main` from GitHub so the PC runs exactly the tested version (the commands are in the chat that came with this guide).
2. **Start the site in Local** and open `http://<your-site>.local/bloodlink/` to check it loads.
3. **Back up your current database.** In Local: your site → **Database** tab → **Open Adminer** → `donor_app` → **Export** → **Execute**. Keep the file — it lets you put everything back after the defence.
4. **Load the demo data.** Still in Adminer with `donor_app` selected: **Import** → choose `db/schema.sql` → **Execute**; then **Import** → `db/seed.sql` → **Execute**. This replaces the data in `donor_app` with the known demo accounts listed in the cheat sheet (Part 6).
5. **Rehearse the whole script once** with a timer, out loud.
6. **Reload the demo data again** (step 4). The rehearsal registered Aisha and used up Kamalu's donation; reloading gives you a clean stage.

ON THE DAY — 30 TO 60 MINUTES BEFORE YOU PRESENT

1. **Reload the demo data one last time** (schema.sql, then seed.sql). Two things in the seed are tied to the time you load it: Zainab is "screened **today**", and Hassan's emergency is due **3 hours** after loading — loading close to the demo keeps both looking right.
2. **Check the internet.** The map tiles, routes and address look-ups come from free online services. Keep a phone hotspot ready.
3. **Open three separate browser sessions** — one browser keeps one login, so each role needs its own window:
   - **Window A — Chrome:** `http://<your-site>.local/bloodlink/` (Home page). You will register Aisha here; stay logged out.
   - **Window B — Chrome Incognito** (Ctrl+Shift+N): sign in as the officer at `user-login.php` — role **Hospital Officer**, `maryam@bloodlink.test` / `Officer@123`. Leave it on the Dashboard.
   - **Window C — Microsoft Edge:** sign in as the admin at `login.php` — `admin` / `admin`. Leave it on the Dashboard.
4. **Make it readable on the projector:** Ctrl and + to zoom each window to 110–125%. Close other tabs and apps; turn on Windows **Focus assist / Do not disturb**; plug in the charger.
5. **Keep the last page of this guide** (the cheat sheet) printed or on your phone.

> TWO RULES THAT SAVE A DEMO
>
> Never register Aisha twice without reloading the seed (the email will already exist). And if a page looks wrong, don't apologise for long: say what *should* happen, use the recovery table in Part 5, and move on — lecturers grade how well you understand your system.

Part 2 / Run sheet

The 18-minute plan at a glance

CLOCK	SEGMENT	WINDOW	THE ONE THING THEY MUST REMEMBER
0:00	1 · Opening	A · Home	The problem, and that you will follow one unit of blood
1:30	2 · How it fits together	A · Home	Browser → PHP → MySQL (12 tables), three roles, free map services
3:00	3 · Register a member	A	Validated, geocoded, hashed password, Member ID, admin notified
4:30	4 · One account does both ✂	A	Donor availability + the online Health Check
5:30	5 · File a Critical request	A	Priority 86 = 60 + 6 + 20; three eligible donors alerted
7:00	6 · Officer: bell and Emergency Board	B	Live bell; board ranks by priority and time left
8:15	7 · Match Donor	B	Only compatible, eligible, non-deferred donors, ranked
9:30	8 · Screen and confirm	B	No donation without a same-day screening; one safe transaction
12:00	9 · A deferral ✂	B	Rules decide; results kept confidential
13:00	10 · The requester is told	A	Fulfilled, with the donor's name
13:45	11 · Nearest blood bank	A	Pinned without a click; shortest road route with km and time
15:15	12 · Administrator ✂	C	Stock per bank, low-stock alerts, reports and CSV, inbox
16:45	13 · Security, testing, close	any	Secure by design; 297 automated checks passed
18:00	**Questions** — about 2 minutes of buffer if you have 20.		

IF YOU ONLY HAVE 15 MINUTES

Skip the segments marked ✂: in 4 say one sentence about availability and the Health Check; in 9 just describe a deferral ("if Grace had said she had malaria recently..."); in 12 show the dashboard and Blood Stock only. That saves about 3 minutes.

Part 3 / The script

What to do and say, step by step

0:00 – 1:30 **1 · Opening** Window A · Home page

SAY

“Good morning. My project is BloodLink, an enhanced blood bank management system.”

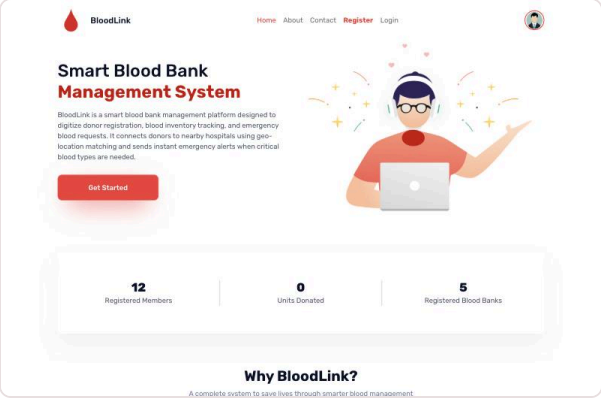
“When a patient needs blood urgently, hospital staff still phone around and search records to find someone who can donate. Three things slow them down: knowing quickly who is compatible and allowed to donate, knowing which blood bank is nearest, and being sure the donor is healthy enough to give.”

“BloodLink brings donors, hospital officers and administrators into one system. It ranks emergencies, matches compatible donors, screens every donor before any blood is taken, and shows the nearest blood bank with the shortest route.”

“In the next few minutes I will follow one unit of blood, from the moment a patient needs it to the moment it is donated.”

POINT AT

the live counters on the Home page — registered members, units donated and registered blood banks come straight from the database.



Home page — the counters are live from the database.

1:30 – 3:00 2 · How it fits together

Window A · Home (or the architecture figure in Part 4)

SAY

“BloodLink is a web application, so it works in any browser, on a computer or a phone.”

“The browser sends requests to PHP pages running on a local web server — Local by Flywheel, with nginx and PHP 8.2. All the data lives in a MySQL database with twelve tables.”

“There are three roles. A member has one account that can both donate and request blood. A hospital officer belongs to one blood bank and does the matching and screening. The administrator runs the network: banks, stock, members, officers and reports.”

“For location we use free, open services instead of paid Google Maps: OpenStreetMap for the map, Nominatim to turn addresses into coordinates, and OSRM to calculate driving routes.”

“Everything turns around one chain: a request comes in, it is ranked, compatible donors are matched, the chosen donor is screened, the donation is recorded, and everyone involved is notified. That is exactly what I will show you now.”

IF THEY ASK

Why PHP and MySQL?

They run on almost any hosting, cost nothing, and suit a form-and-database system like this; Local by Flywheel gives the same nginx/PHP/MySQL stack on a laptop.

3:00 – 4:30 3 · Register a member

Window A · Register

DO

1. Click **Register** in the top menu.
2. Full Name `Aisha Mohammed` · Email `aisha@bloodlink.test` · Phone `08031234567` · Blood group `A+` · leave Member ID empty · Address `Sabon Tasha, Kaduna` · Password `Demo@2026` twice.
3. Click **Submit Registration** → the welcome card shows her Member ID.

SAY

"Let me register a new member, Aisha."

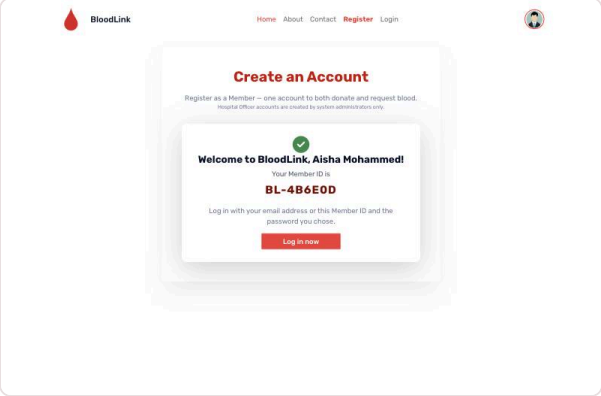
"Every field is checked on the server, not just in the browser: a valid email and phone number, a real blood group, and a password of at least eight characters with a letter and a number. Duplicate emails and Member IDs are refused."

"When I save, her address is sent to Nominatim and stored as latitude and longitude, so the system can later work out how far she is from any hospital. Her password is never stored — only a bcrypt hash."

"She gets a Member ID, and the administrator receives a "New member registered" notification."

POINT AT

the generated Member ID on the welcome card.



The welcome card with the generated Member ID.

IF THEY ASK

What if the address can't be found?

The member is still saved; they just don't appear on the map until they correct the address in My Profile.

4:30 – 5:30 4 · One account does both ✂

Window A · Member dashboard

DO

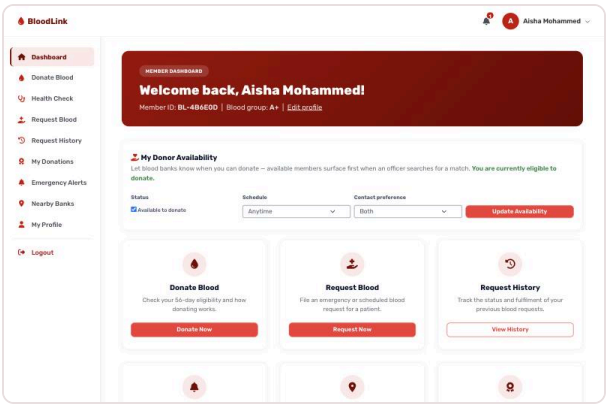
1. Click **Log in now** → role **Member (Donor / Recipient)** → `aisha@bloodlink.test` / `Demo@2026` → **Login**.
2. Point at **My Donor Availability**, then at **Health Check** in the sidebar (no need to open it).

SAY

"I will log in as Aisha, with her email or her Member ID."

"One account is both donor and recipient. Here she sets whether she is available to donate, when, and how she prefers to be contacted — available donors are offered to officers first."

"In the menu there is a Health Check: a one-minute questionnaire a member can answer before going to the bank. It tells them straight away if anything would stop them donating. I will show how the officer uses it in a moment."



Member dashboard — availability, eligibility, and every member action in one menu.

5:30 – 7:00 5 · File a Critical request

Window A · Request Blood

DO

1. Click **Request Blood**.
2. Choose **Critical Emergency**; Needed within (hours) `2`.
3. Patient `Ibrahim Yusuf` · Blood group `O+` · Units `1` · Hospital: start typing `Barau` and pick **Barau Dikko Teaching Hospital** · Location `Kaduna` · Phone `08031234567`.
4. Click **Submit Request** → "Your blood request has been submitted successfully."

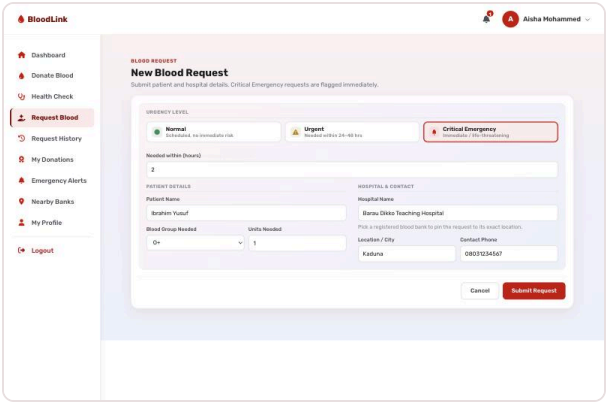
SAY

"Aisha's relative, Ibrahim Yusuf, urgently needs one unit of O-positive blood at Barau Dikko Teaching Hospital within two hours."

"Three things happen the moment she submits. First, the request gets a priority score out of 100: critical severity gives 60, the rarity of O-positive adds 6, and less than three hours left adds 20 — so 86."

"Second, because it is a Critical Emergency, an emergency alert is posted and every member who could donate right now — compatible, outside the 56-day recovery period and not deferred — is notified directly. Here that is three people."

"Third, every hospital officer and the administrator are notified. And because she chose a registered blood bank as the hospital, the request is pinned to that bank's exact location, which matters for distances."



The request form with Critical Emergency selected.

IF THEY ASK

How is the priority calculated?

severity (critical 60 · severe 40 · moderate 20) + rarity × 2 (AB− rarest = 20 … O+ commonest = 6) + time left (≤ 3 h: 20 · ≤ 12 h: 12 · ≤ 48 h: 6), capped at 100 and recalculated whenever the board is opened.

7:00 – 8:15 6 · The officer: bell and Emergency Board

Window B · Officer (Maryam)

DO

1. Reload the page (F5) if the badge hasn't updated; click the **bell** to open it.
2. Click **Emergency Board** in the sidebar.

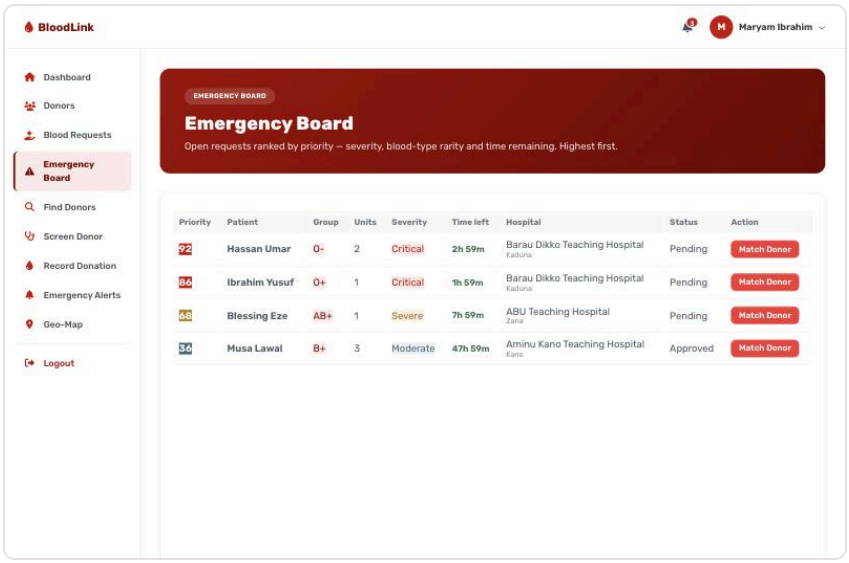
SAY

"Now I am Maryam, a hospital officer at Barau Dikko. Her bell already shows "New O+ request" — the bell checks for new notifications every 30 seconds without reloading the page."

"The Emergency Board ranks every open request by priority, with a countdown to its deadline. Notice Hassan's O-negative request is above ours: it has less time left, so its priority has risen to 92. The board recalculates priorities every time it opens, so a request climbs as its deadline gets closer."

POINT AT

the red priority numbers and the "Time left" column.



Emergency Board — highest priority first, with time left.

IF THEY ASK

Why is Hassan first?

His deadline was set when the demo data was loaded, so he now has less time left; his O− blood is also rarer than O+. (If his deadline has passed, say so — it shows the board counts down in real time.)

8:15 – 9:30 7 · Match Donor

DO

1. On the board, click **Match Donor** next to **Ibrahim Yusuf.**

SAY

“BloodLink only offers donors who are compatible with O-positive, have not donated in the last 56 days, are not deferred by a health screening — and never the person who asked.”

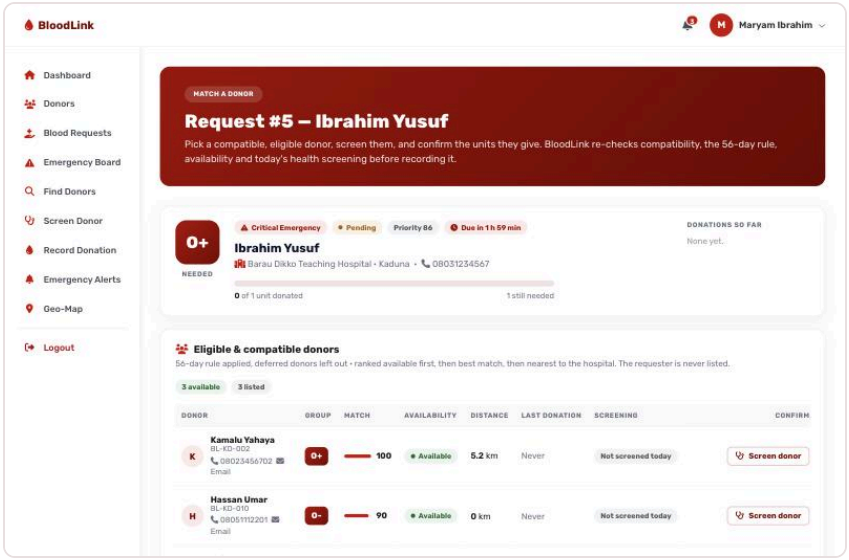
“Kamalu is O-positive, an exact match, so he scores 100. Zainab and Hassan are O-negative, the universal donor, so they score 90. Available donors come first, then the best match, then whoever is nearest to the hospital — these distances are measured from the hospital's coordinates.”

“Grace is also O-positive, but she isn't listed: her own Health Check said she had malaria recently, so she is deferred.”

“And look at this column: there is no Confirm button for Kamalu yet, only "Screen donor". Nobody can donate without passing a health screening the same day.”

POINT AT

the match score bars, the Distance column and the Screening column.



Match Donor — compatible, eligible donors ranked; unscreened donors can only be screened.

IF THEY ASK

How do you know who is compatible?

A standard ABO/Rh table (Part 4): an O+ patient can receive O+ or O−. The server checks it again when the donation is confirmed, so it cannot be bypassed from the browser.

9:30 – 12:00 8 · Screen the donor and confirm the donation

Window B · Screen Donor → Match Donor

DO

1. Click **Screen donor** next to **Kamalu Yahaya**.
2. Point at "Pre-filled from the member's self-check".
3. Vital signs: Weight `72` · Haemoglobin `14.2` · BP `122` / `78` · Pulse `74` · Temperature `36.6`.
4. Rapid tests: click **Negative** for all five → **Save screening**.
5. Back on Match Donor ("Kamalu Yahaya passed today's health screening"): leave `1` unit → **Confirm** → **OK**.
6. "Donation recorded (1 unit(s) of O+). Request is now Fulfilled." — point at the full progress bar and "Donations so far".

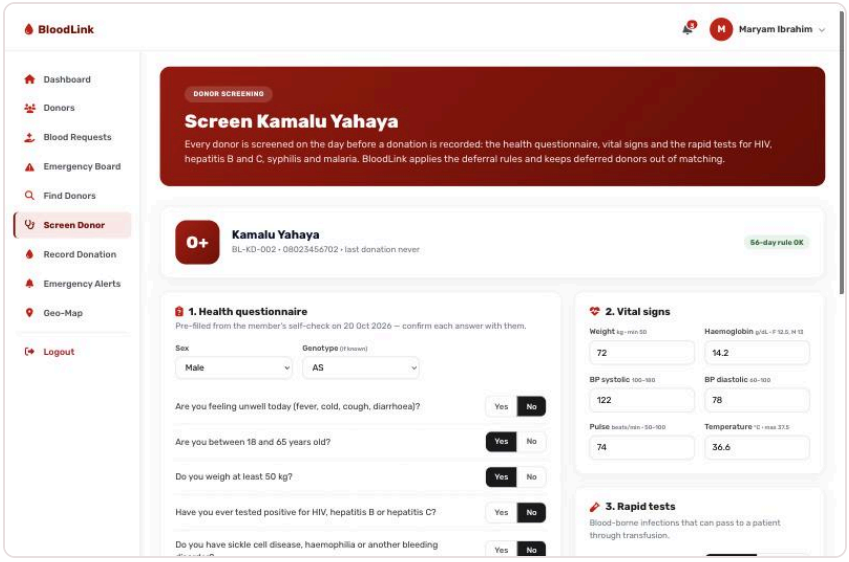
SAY

"The questionnaire is already filled in from the Health Check Kamalu did online, so the officer simply confirms each answer with him."

"Then the vital signs: weight 72 kilograms, haemoglobin 14.2, blood pressure 122 over 78, pulse 74, temperature 36.6. Then the five rapid tests for infections that can be passed on through blood — HIV, hepatitis B, hepatitis C, syphilis and malaria. All negative."

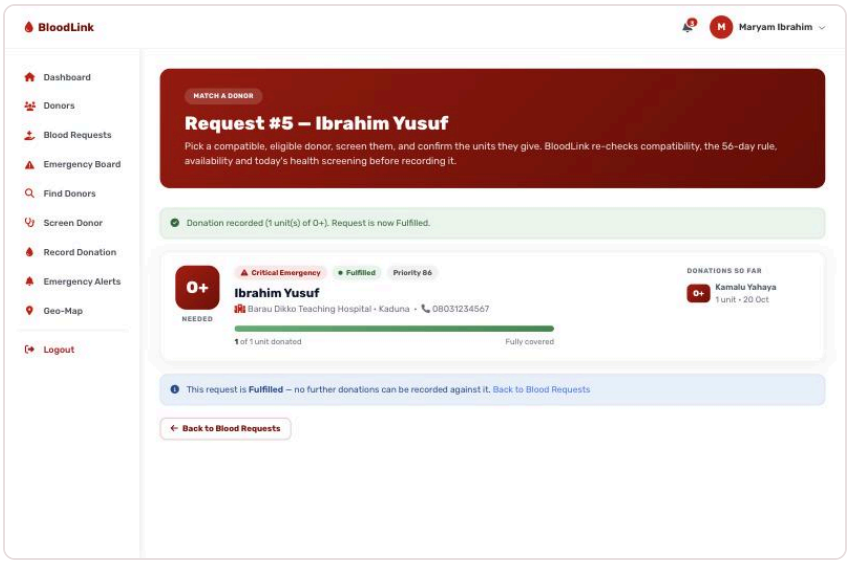
"When I save, BloodLink's rules check everything: at least 50 kilograms, haemoglobin at least 13 for men and 12.5 for women, blood pressure and pulse in range, no fever, no reactive tests. He is eligible, and it brings me straight back to the request — where Kamalu now has a Confirm button."

"Confirming runs as one database transaction. It locks the donor and the request so two officers can never record the same donor twice, re-checks compatibility, the 56-day rule and today's screening, records the donation under the donor's own blood group and links it to the screening, resets his 56-day clock and marks the request Fulfilled. Blood given for a named patient goes to that patient, so the bank's stock doesn't change; a walk-in donation would add to it."



Screen Donor — the questionnaire pre-filled from Kamalu's Health Check, vital signs and rapid tests.

> **IF THEY ASK**
>
> **What if the officer leaves something out?**
>
> The screening is refused ("Enter the weight...") and nothing is saved.
>
> **Can a screening be reused?**
>
> No — one screening clears one donation, on the day it was done.



After Confirm: the request is Fulfilled and Kamalu appears under Donations so far.

12:00 – 13:00 9 · A deferral ✂

Window B · Screen Donor

DO

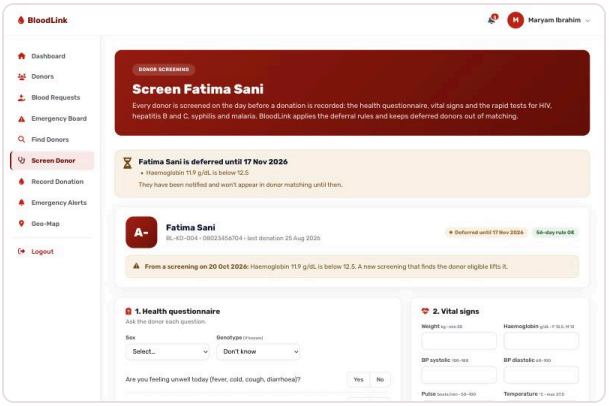
1. Sidebar → **Screen Donor** → type `BL-KD-009` → **Start screening**.
2. Vital signs: `63` · `13.0` · `120` / `80` · `76` · `36.8` ; all five tests **Negative** → **Save screening**.
3. "Grace Okonkwo is deferred until ..." — "Malaria in the last 4 weeks".

SAY

"Here is what happens when a donor should not give blood. This is Grace — her answers are pre-filled again, and she said yes to malaria in the last four weeks."

"Her vital signs and tests are fine, but BloodLink defers her for 28 days and tells her the date she can donate again. Other examples: low haemoglobin, 28 days; high blood pressure, 7 days; a recent tattoo, six months; a reactive HIV or hepatitis test is a permanent deferral."

"Screening results are confidential: when an infection test is reactive, the member's notification never names the test — it only asks them to speak to the bank in private."



A temporary deferral, with the reason and the date it ends (shown here for low haemoglobin).

13:00 – 13:45 10 · The requester is told

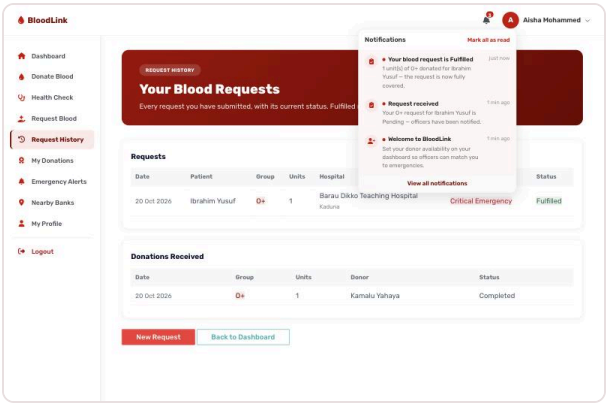
Window A · Aisha

DO

1. Window A: reload the page; click the **bell** → "Your blood request is Fulfilled".
2. Sidebar → **Request History** → status **Fulfilled**; **Donations Received** shows Kamalu Yahaya.

SAY

"Back to Aisha. Her bell says her request is fulfilled, and Request History shows who donated. Every step in this story sent the right people a notification — about a dozen messages, without anyone making a phone call."



The requester's bell and Request History.

13:45 – 15:15

11 · The nearest blood bank

Window A · Nearby Banks

DO

1. Sidebar → **Nearby Banks**; wait for the route.
2. Drag the blue pin a little, then click a different bank in the list.
3. Open **Turn-by-turn directions**.

SAY

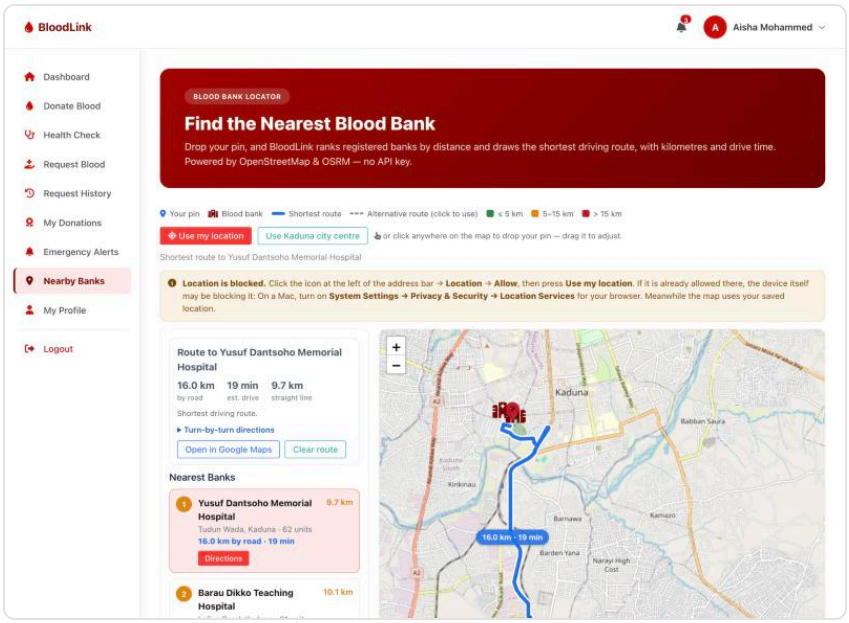
"Finally, Aisha needs to get to a blood bank. Without clicking anything, she is pinned at her registered address in Sabon Tasha."

"The banks are ranked by straight-line distance using the Haversine formula, and the shortest driving route to the nearest one is drawn in blue, with kilometres and drive time from OSRM. Alternative routes are grey, and there are turn-by-turn directions."

"If she drags the pin or picks another bank, it re-routes. Live GPS is used when the site is served securely over HTTPS — browsers only share GPS with secure pages, so on this local server it uses her saved address and says so."

POINT AT

the blue route label ("x km · y min") and the ranked list on the left.



Blood Bank Locator — pinned at the saved address, shortest route drawn.

IF THEY ASK

Why not Google Maps?

Google requires a billing account; OpenStreetMap, Nominatim and OSRM are free and open.

What if the routing service is down?

The map falls back to a straight line with its distance.

15:15 – 16:45

12 · The administrator ✂

Window C · Admin

DO

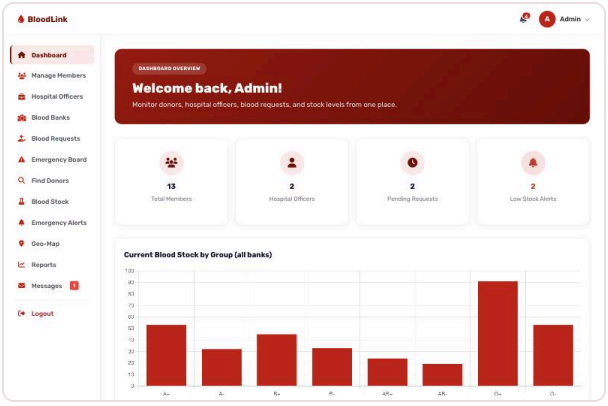
1. Window C: reload the **Dashboard** — point at the counters and the chart.
2. Sidebar → **Blood Stock** → choose **Barau Dikko** → point at the highlighted groups.
3. Sidebar → **Reports** → **Export CSV Report**.
4. Sidebar → **Messages** (one message from Dr. Bala Usman).

SAY

"The administrator sees the whole network: members, officers, open requests, active low-stock alerts, and stock by blood group across all banks."

"Stock is kept per blood bank. At Barau Dikko, A-negative and O-negative are below their thresholds, so low-stock alerts are active — when a group falls below its threshold, that bank's officers and the administrators are notified once, and the alert clears automatically when stock recovers."

"Reports can be filtered by date and exported as a CSV file, and messages sent from the public Contact page arrive in this inbox."



Admin dashboard — network counters and stock by blood group.

16:45 – 18:00

13 · Security, testing and close

any window

SAY

“Security was designed in, not added later: passwords are hashed with bcrypt, every database query uses prepared statements against SQL injection, every form carries a token against forged submissions, the session is renewed at each login, and every page checks the user's role.”

“I tested it in two ways: manual walkthroughs of every role, and automated test scripts — 100 regression checks, 35 notification checks, and a crawl that loads every page for every role, 162 page loads. All of them pass.”

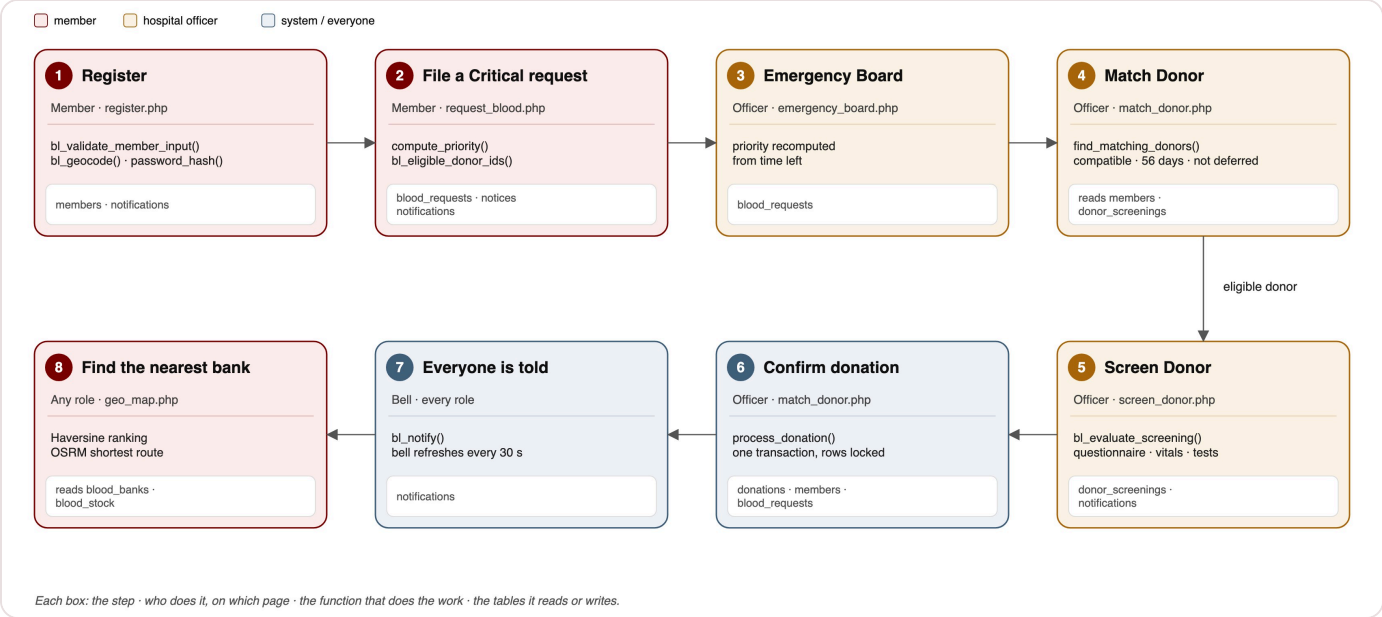
“The limitations I would address next: notifications are in-app only, so SMS or e-mail alerts would reach donors faster; it should be deployed on a secure HTTPS server, which would also enable live GPS; and the rapid tests are a first screen, not a replacement for laboratory testing.”

“BloodLink turns a slow, manual search into a few minutes of guided steps: the right donor, screened and safe, at the nearest bank. Thank you — I'm happy to take your questions.”

Part 4 / How it connects

Underneath the demo

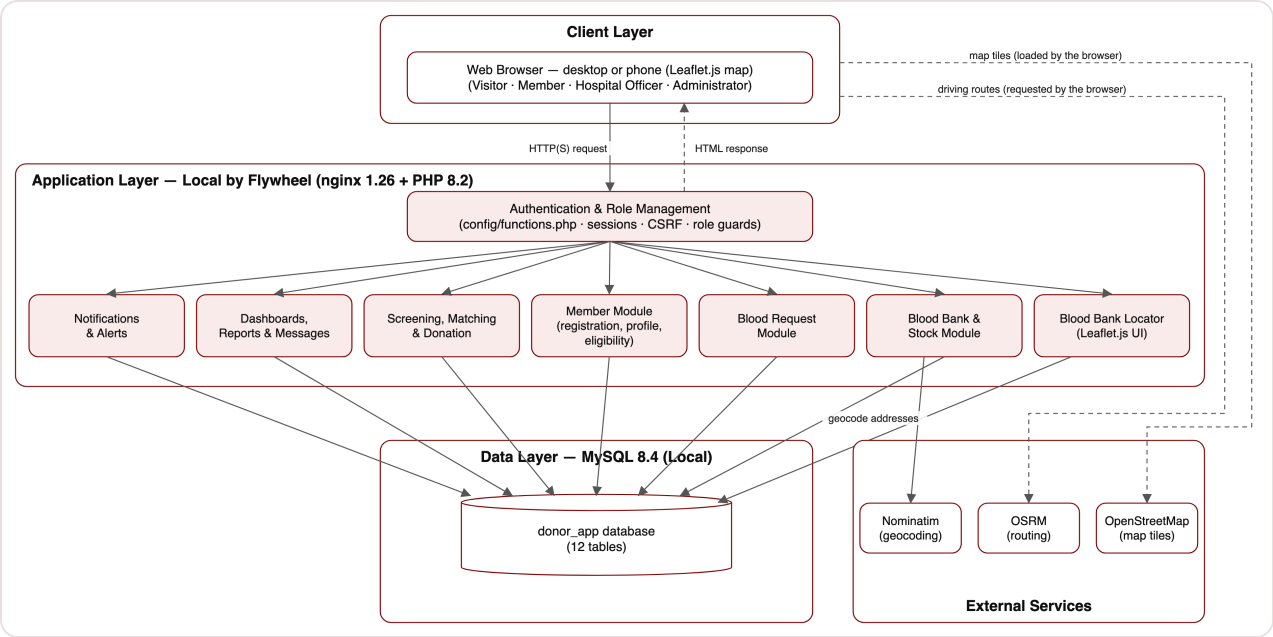
Use this section to answer "how does it work?" with confidence. The figure follows the demo; the table says exactly what each step reads, writes and announces.



The demo as one chain — each box names the page, the function that does the work, and the tables it touches.

STEP	WHAT THE CODE DOES	SAVED IN	WHO IS NOTIFIED
Register	Validates every field (`bl_validate_member_input`), geocodes the address with Nominatim (`bl_geocode`), hashes the password, generates a unique Member ID.	members	Admins: "New member registered"; the member: "Welcome to BloodLink"
File a request	Maps urgency to severity, computes the priority (`compute_priority`), uses the bank's coordinates when the hospital is a registered bank.	blood_requests; for Critical also notices	Requester; every officer; admins; for Critical, every eligible compatible member (`bl_eligible_donor_ids`)
Emergency Board	Recomputes each open request's priority from the time left and sorts highest first.	blood_requests.priority_score	—
Match Donor	`find_matching_donors`: compatible groups only, outside 56 days, not deferred (`bl_deferred_sql`), requester excluded; ranked available → match score → distance (Haversine in SQL) → longest since donating.	reads members, donor_screenings	—
Self-check	The member's questionnaire is evaluated by `bl_evaluate_screening` (answers only).	donor_screenings (source self)	—
Screen Donor	`bl_evaluate_screening`: refuses incomplete screenings; applies the deferral rules to the answers, vital signs and five tests; the longest deferral wins.	donor_screenings (source officer)	The donor — never naming an infection test
Confirm	`process_donation` in one transaction with the donor and request rows locked (`SELECT … FOR UPDATE`): re-checks units (1–2), blood group, compatibility, 56 days, availability, deferral and today's screening; records the donation; resets the donor's clock; sets Approved / Fulfilled.	donations (+ screening_id), members.last_donation_date, blood_requests.status	Donor: "Thank you"; requester: "Approved / Fulfilled"

Walk-in	Same `process_donation` without a request: units are added to the officer's bank stock and its low-stock alert is re-checked (`bl_sync_stock_alert`).	donations, blood_stock, stock_alerts	Donor; if stock falls below threshold, that bank's officers and admins (once)
Locator	Starts at the saved address (member) or own bank (officer); asks for GPS on secure pages; ranks banks with Haversine; asks OSRM for up to three driving routes and draws the shortest.	reads blood_banks, blood_stock	—



Architecture (the defence document's Figure 5.1): client, application and data layers, and the external map services.

WHO CAN RECEIVE FROM WHOM (GET_COMPATIBLE_DONOR_TYPES)

PATIENT	CAN RECEIVE FROM	PATIENT	CAN RECEIVE FROM
O−	O−	O+	O+, O−
A−	A−, O−	A+	A+, A−, O+, O−
B−	B−, O−	B+	B+, B−, O+, O−
AB−	AB−, A−, B−, O−	AB+	everyone (universal recipient)

Match score: 100 same group · 90 O− (universal donor) · 80 same ABO letter · 70 other compatible group · 0 incompatible (never listed, and refused on confirmation).

THE TWELVE TABLES

TABLE	HOLDS
login	Administrator accounts
blood_banks	The registered facilities, with coordinates — the centre of the design
hospital_workers	Hospital officers; each belongs to one blood bank
members	One account per person: blood group, 56-day clock, address and coordinates, availability
blood_requests	Requests with urgency, deadline, priority score, status, and the member who asked
donor_screenings	Self-checks and officer screenings: answers, vital signs, five test results, outcome and deferral date
donations	The ledger: donor, recipient, request, officer, bank, units, and the screening that cleared it
blood_stock · stock_alerts	Units per bank per blood group; alerts when a group falls below its threshold
notices	Emergency alerts shown to members
notifications	Every bell item, for every role
contact_messages	Messages from the public Contact page

Foreign keys are enforced by MySQL with deliberate delete rules — for example, deleting a blood bank removes its stock rows (CASCADE) but keeps its donations and officers, just clearing the link (SET NULL). On top of that, the app refuses to delete a member who has donated or an officer who recorded donations, so the ledger is never broken. The full ERD and data dictionary are in the defence document (Figure 5.3 and Appendix A).

SCREENING RULES IN ONE LINE EACH

- **Permanent:** reactive HIV, hepatitis B or C; past positive HIV/hepatitis; sickle cell disease or a bleeding disorder (or genotype SS/SC); serious chronic illness; injected non-prescribed drugs.
- **365 days:** reactive syphilis; transfusion or contact with HIV/hepatitis in 12 months; outside 18–65. **182 days:** pregnancy or recent birth/breastfeeding; tattoo, piercing or acupuncture; major surgery.
- **90 days:** under 50 kg. **28 days:** malaria; haemoglobin under 12.5 (women) / 13.0 (men). **14 days:** unwell, or temperature over 37.5 °C. **7 days:** antibiotics; blood pressure outside 100–180 / 60–100; pulse outside 50–100.

Likely questions, with answers

What problem does BloodLink solve that a spreadsheet can't?

It acts on the data: it ranks emergencies automatically, offers only donors who are compatible, eligible and healthy, refuses unsafe donations on the server, notifies the right people at each step, and finds the nearest bank with a route. A spreadsheet only stores names.

How do you enforce the 56-day rule?

Each member has a last donation date. The matching query leaves out anyone inside 56 days, and when an officer confirms, `process_donation` checks it again inside the transaction, so it can't be bypassed by an old page or a modified form.

What if two officers confirm the same donor at the same moment?

The confirmation runs in one transaction that locks the donor's and the request's rows (`SELECT … FOR UPDATE`). The second officer waits, then sees that the donor has just donated (56-day rule) or that the request is already fulfilled, and is refused. A screening can also clear only one donation.

Can the system tell whether a donor has a disease?

It doesn't diagnose. It records the results of the rapid tests done at the bank and the donor's answers, and applies standard donor-selection rules to decide whether they may donate. Confirmatory laboratory testing of the collected blood remains outside the system.

How do you protect sensitive medical data?

Only officers see screening details; members see only their own outcome, and notifications never name an infection test. Passwords are bcrypt-hashed, queries use prepared statements, forms carry CSRF tokens, sessions are renewed at login, and every page checks the role. In production it must run over HTTPS on a secured server.

How does the map find the shortest route?

Banks are first ranked by straight-line distance (Haversine). For the chosen bank, the browser asks OSRM for up to three driving routes along real roads; BloodLink draws the shortest by distance in blue and the others in grey, with kilometres and estimated drive time.

Where does the user's location come from?

Addresses are converted to coordinates with Nominatim when members, banks and requests are saved. On the map a member starts at that saved address and an officer at their bank; live GPS is used when the page is secure (HTTPS or localhost), because browsers refuse to share GPS with plain http pages.

Why didn't you use Google Maps?

Google requires a billing account. OpenStreetMap, Nominatim and OSRM are free and open-source, and Leaflet displays the map.

What happens without internet?

Everything in the database still works: registration, requests, matching, screening and donations. The map tiles, address look-ups and road routes need internet; if routing fails the map shows a straight line with its distance.

How is the priority score calculated, and why?

Severity (critical 60, severe 40, moderate 20) plus blood-type rarity × 2 (AB− highest) plus time pressure (up to 20 when three hours or less remain), capped at 100. It puts life-threatening, rare and nearly-due requests first, and it is recalculated whenever the board is opened.

Why one "member" account instead of separate donors and recipients?

Real people both give and need blood. One account avoids duplicate records, lets the same person donate and request, and keeps their blood group and 56-day clock in one place. It also lets the system exclude a requester from donating to their own request.

When does blood stock change?

A walk-in donation adds units to the officer's bank stock. A donation for a named patient goes directly to that patient, so stock is unchanged. Administrators can also update stock; whenever a group falls below its threshold an alert is raised and notified once, and it clears when stock recovers.

How did you test it?

Manual walkthroughs for every role, plus automated scripts against a local server with the same PHP and MySQL versions: 100 regression checks (registration, login, access control, screening, matching and donation rules, stock and alerts), 35 notification checks, and a crawl of every page for every role (162 page loads). The seed data deliberately places donors on both sides of the 56-day boundary.

What bugs did testing find?

For example: donations were recorded under the request's blood group instead of the donor's; the server accepted incompatible donors and negative unit counts; duplicate Member IDs blocked logins; and requests typed as free text were placed kilometres from their hospital. Each was reproduced, fixed and re-tested.

How would it scale to the whole country?

The design already keeps stock, officers and alerts per blood bank, and more banks are simply added. A national deployment would need proper hosting with HTTPS, its own routing server instead of the public OSRM demo, SMS alerts, and backups — not a redesign.

What would you add next?

SMS and e-mail alerts, an HTTPS deployment (which also enables GPS), a mobile app for donors, integration with laboratory systems for confirmatory test results, and analytics on donor response times.

Why PHP without a framework?

For a final-year project it keeps every step visible and explainable, and runs on any PHP host. Shared logic lives in one operations class and helper functions, so a framework could be introduced later

without changing the database.

What does the administrator do that an officer can't?
Manage members, officers and blood banks, set stock and thresholds for any bank, see reports across the network and export them, and read contact messages. Officers work their own bank: matching, screening and recording donations.

If something goes wrong

WHAT YOU SEE	WHAT TO DO / SAY
"Something went wrong" on every page	The database isn't running — start the site in Local, then reload. Say: "the database service stopped; BloodLink shows a safe error page instead of technical details."
Grey map with no streets	No internet for the map tiles. Switch to the hotspot and reload; meanwhile point at the ranked list and distances, which are calculated by BloodLink itself.
Route shows "straight line"	The free OSRM server didn't answer. Say: "this is the fallback — when routing is unavailable it still shows the straight-line distance." Try again in a moment.
Amber note about GPS on the map	Expected on http — turn it into a point: "browsers only share GPS with secure pages; deployed on HTTPS this uses live GPS."
"Email already registered" for Aisha	You rehearsed without reloading the seed. Use `aisha2@bloodlink.test` and carry on.
Login refused	Check the role dropdown (Member vs Hospital Officer) and the password; members can use their email or Member ID.
No Confirm button for a donor	That is the screening rule — click **Screen donor** first.
Kamalu or Grace missing from Match Donor	The data wasn't reloaded after the rehearsal (Kamalu has already donated or Grace is already deferred). Use Hassan instead: screen him and confirm.
Zainab not "Passed today"	The seed was loaded on an earlier day — nothing breaks; you screen Kamalu anyway.
Bell badge didn't change	It refreshes every 30 seconds — press F5.
Text too small on the projector	Ctrl and + to zoom in; Ctrl and 0 to reset.

Part 6 / Cheat sheet

Keep this page next to you

WHAT	VALUE
Site	`http://<your-site>.local/bloodlink/`
Admin (Edge)	login.php · admin / admin
Officer (Incognito)	user-login.php · Hospital Officer · maryam@bloodlink.test / Officer@123
New member (Chrome)	Aisha Mohammed · aisha@bloodlink.test · 08031234567 · A+ · Sabon Tasha, Kaduna · Demo@2026
Request	Critical Emergency · 2 hours · Ibrahim Yusuf · O+ · 1 unit · Barau Dikko Teaching Hospital · Kaduna · 08031234567
Expect	Priority 86 · 3 donors alerted · Kamalu 100, Zainab and Hassan 90 · Grace not listed
Screen Kamalu	72 kg · Hb 14.2 · BP 122/78 · pulse 74 · 36.6 °C · all five tests Negative → Confirm 1 unit
Defer Grace	BL-KD-009 · 63 kg · Hb 13.0 · BP 120/80 · pulse 76 · 36.8 °C · tests Negative → deferred 28 days (malaria)
Other members	Member@123 — zainab@ (O−, screened today) · hassan@ (O−, own emergency) · kamalu@ · grace@ · all @bloodlink.test

THE SIX SENTENCES TO LAND, WHATEVER HAPPENS

1. One account lets a person both donate and request blood.
2. Emergencies are ranked automatically by severity, rarity and time left.
3. Only compatible, eligible, healthy donors are offered — and the server re-checks it.
4. No donation is recorded without a same-day health screening.
5. Everyone involved is notified at every step, confidentially where it matters.
6. The nearest bank and the shortest route are shown without a click.

AFTER THE DEFENCE

To get your previous data back: Local → Database → Open Adminer → `donor_app` → Import → the backup you exported the night before → Execute.